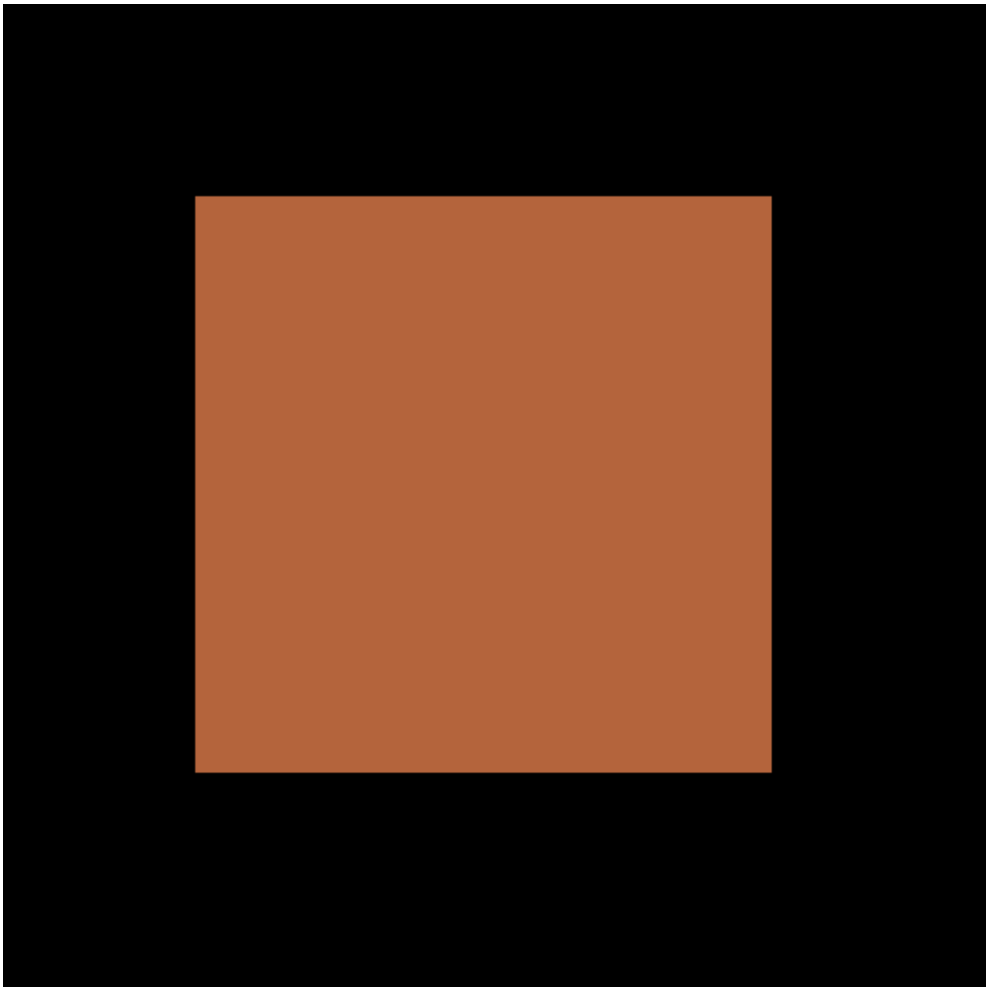


# Aakhi — Retinal Image Analysis Report

Phase 1 Report (DR, OD/OC, Lesions, Glaucoma)

|              |              |              |                              |
|--------------|--------------|--------------|------------------------------|
| Patient Name | Test Patient | Doctor       | Dr. Test Doctor              |
| Age          | 45           | Date         | 08 May 2026, 10:12 AM        |
| Gender       | Male         | Eye Examined | Right Eye (OD)               |
| Report ID    | A80EB758     | Institution  | IIT Bhubaneswar — Eye AI Lab |

## Fundus Image



Original fundus photograph

## DR Grading

Classification: **Mild DR** (Level 1/4)

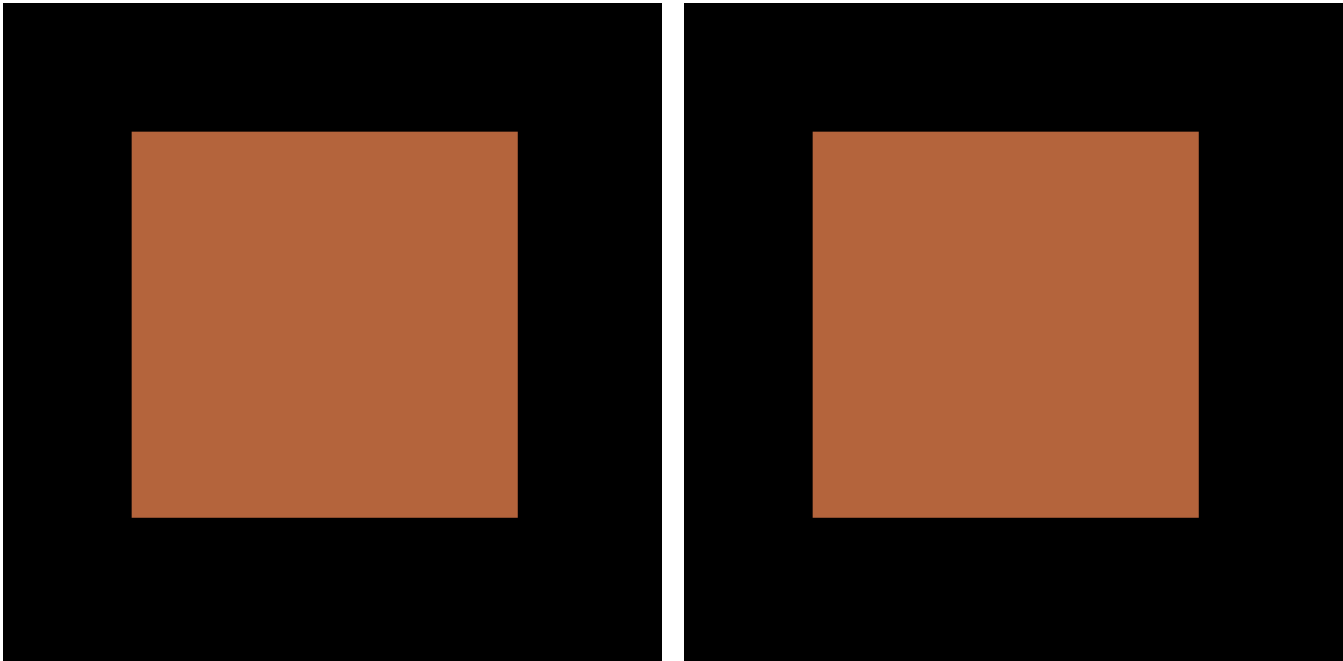
Clinical Description: Mild non-proliferative DR. Presence of at least one microaneurysm. Regular monitoring recommended.

|                   |                                                          |
|-------------------|----------------------------------------------------------|
| DR Severity Level | 1 / 4 — Mild DR                                          |
| Grading Scale     | ETDRS / International Clinical DR Disease Severity Scale |
| Model             | EfficientNet-B6 (fine-tuned, 5-class)                    |

|                    |                                                                          |
|--------------------|--------------------------------------------------------------------------|
| Recommended Action | Follow-up in 6–12 months. Optimise glycaemic and blood pressure control. |
|--------------------|--------------------------------------------------------------------------|

Optic Disc / Cup Segmentation

Color Key: Green=Optic Disc | Blue=Optic Cup | Quadrant Analysis (S=Superior, I=Inferior, N=Nasal, T=Temporal)



Segmentation (Green=Disc, Blue=Cup)                      Overlay with quadrant lines (S/I/N/T)

Morphometric Measurements

|                           |                                                                                            |
|---------------------------|--------------------------------------------------------------------------------------------|
| Vertical CDR (vCDR)       | 0.450 — CDR within normal limits (< 0.5). Low suspicion for glaucomatous optic neuropathy. |
| Horizontal CDR (hCDR)     | 0.400                                                                                      |
| Area-based CDR            | 0.380                                                                                      |
| Optic Disc — Vertical Ø   | 400 px                                                                                     |
| Optic Disc — Horizontal Ø | 420 px                                                                                     |
| Optic Cup — Vertical Ø    | 180 px                                                                                     |
| Optic Cup — Horizontal Ø  | 168 px                                                                                     |
| Disc Area (px²)           | 125664                                                                                     |
| Cup Area (px²)            | 25447                                                                                      |

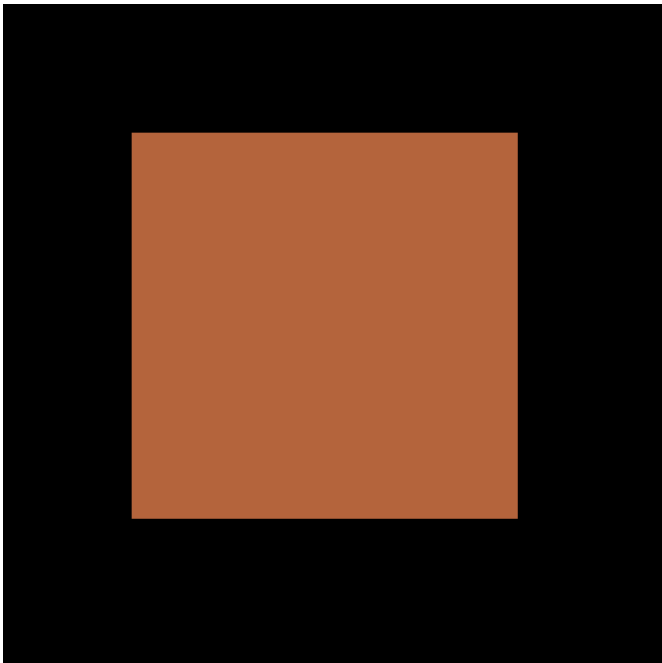
Neuroretinal Rim Analysis — ISNT Rule

Normal ISNT rule: Inferior rim Superior Nasal Temporal

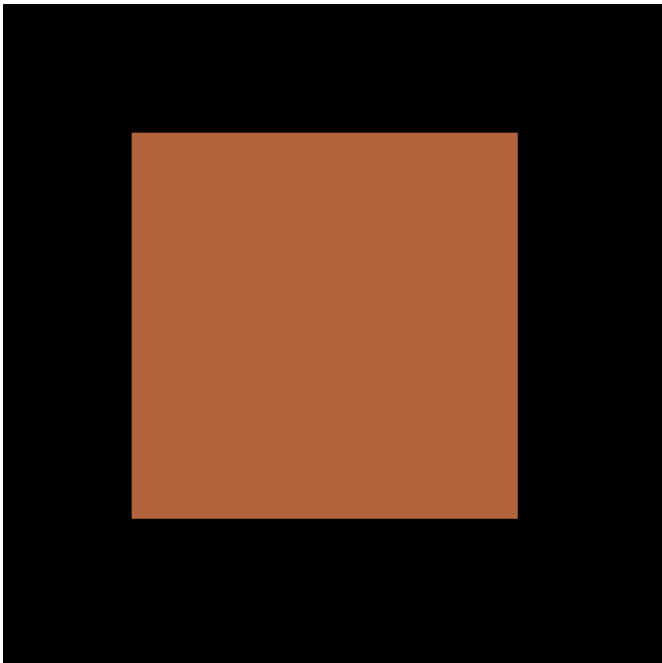
|              |                              |
|--------------|------------------------------|
| Superior Rim | 28.2% of total rim area      |
| Inferior Rim | 29.1% of total rim area      |
| Nasal Rim    | 21.4% of total rim area      |
| Temporal Rim | 21.3% of total rim area      |
| ISNT Rule    | Normal (ISNT rule satisfied) |

Lesion Detection

Color Key: Red=Hard Exudates | Green=Hemorrhages | Blue=Microaneurysms | Yellow=Soft Exudates



Lesion segmentation (colour-coded by type)



Overlay on original fundus

Detected Lesion Areas (pixels<sup>2</sup>)

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| Hard Exudates (HE)  | 500 px <sup>2</sup> — Lipid deposits from leaking vessels |
| Haemorrhages (HEM)  | 200 px <sup>2</sup> — Intraretinal bleeding               |
| Microaneurysms (MA) | 100 px <sup>2</sup> — Small capillary bulges              |
| Soft Exudates (SE)  | 50 px <sup>2</sup> — Cotton-wool spots                    |
| Model               | UNet + FIAM attention (IDRiD dataset)                     |

## Glaucoma Grading

Glaucoma Severity Grade: No Glaucoma

Clinical Interpretation: No evidence of glaucomatous optic neuropathy.

|                     |                                          |
|---------------------|------------------------------------------|
| Model               | SE-ResNet101 (3-class glaucoma grading)  |
| Classification      | No Glaucoma                              |
| Corroborating CDR   | vCDR = 0.450 (suspicion threshold: 0.65) |
| ISNT Rule Violation | No                                       |

## Clinical Summary

|                  |                                             |
|------------------|---------------------------------------------|
| DR Grading       | Mild DR (Level 1/4)                         |
| Glaucoma Grading | No Glaucoma                                 |
| Vertical CDR     | 0.450 — Risk: Low                           |
| ISNT Rule        | Normal                                      |
| Lesion Load      | HE: 500   HEM: 200   SE: 50 px <sup>2</sup> |

## Recommendations

- DR follow-up: Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

Report ID: A80EB758 | 08 May 2026, 10:12 AM

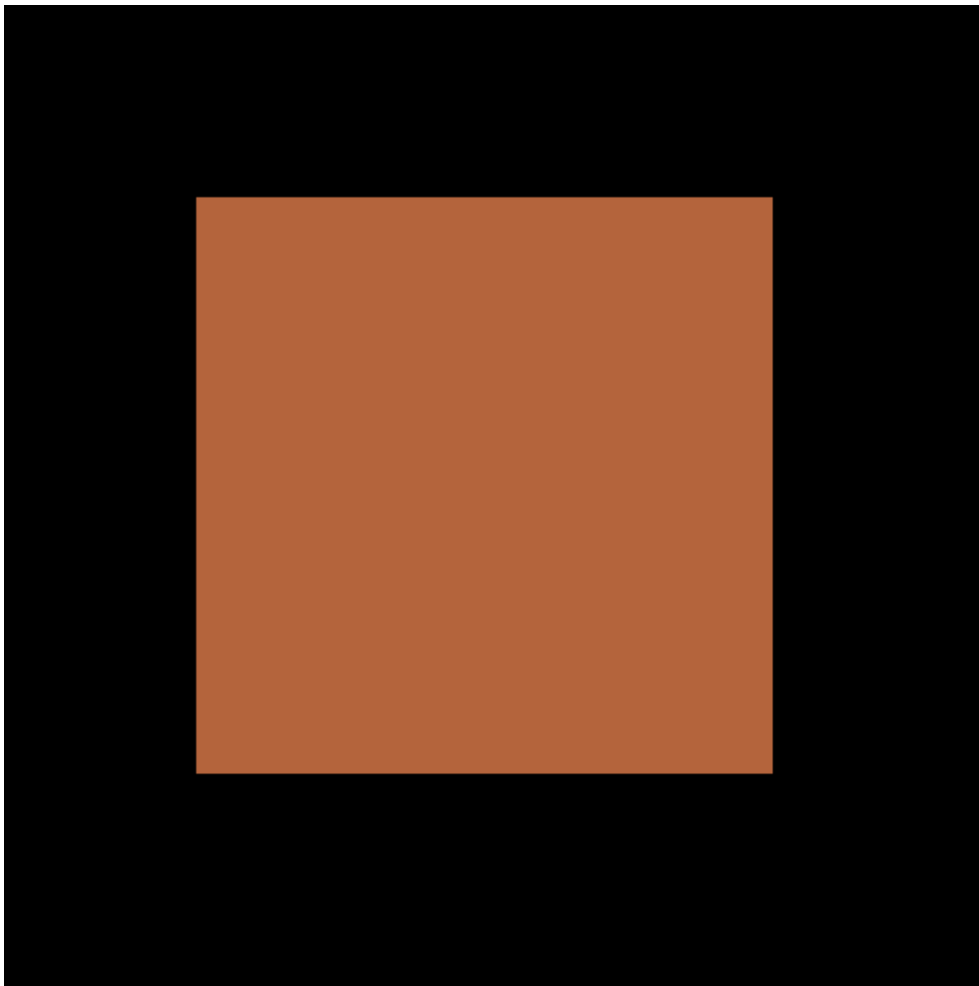
Generated by Aakhi Retinal AI

This report is generated by the Aakhi AI system for clinical reference only. All findings must be verified by a qualified ophthalmologist before treatment decisions.

# ஆகி — விழித்திரை படம் பகுப்பாய்வு அறிக்கை

கட்டம் 1 அறிக்கை (, /, புண்கள், கண் அழத்தம்)

|                  |              |                      |                              |
|------------------|--------------|----------------------|------------------------------|
| நோயாளியின் பெயர் | Test Patient | மருத்துவர்           | Dr. Test Doctor              |
| வயது             | 45           | தேதி                 | 08 May 2026, 10:12 AM        |
| பாலினம்          | Male         | பரிசோதிக்கப்பட்ட கண் | Right Eye (OD)               |
| அறிக்கை          | A80EB758     |                      | IIT Bhubaneswar — Eye AI Lab |



Original fundus photograph

நீரிழிவா விழித்திரை நோய் தரவரிசை

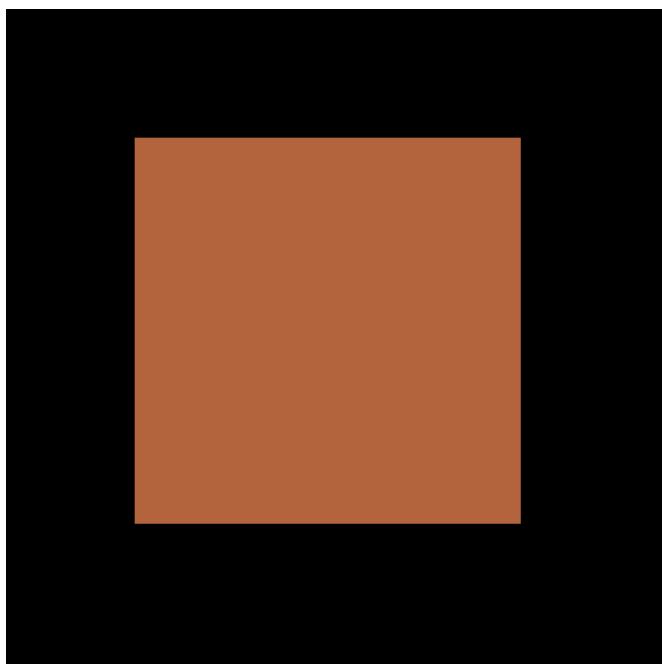
Classification: Mild DR (Level 1/4)

Clinical Description: Mild non-proliferative DR. Presence of at least one microaneurysm. Regular monitoring recommended.

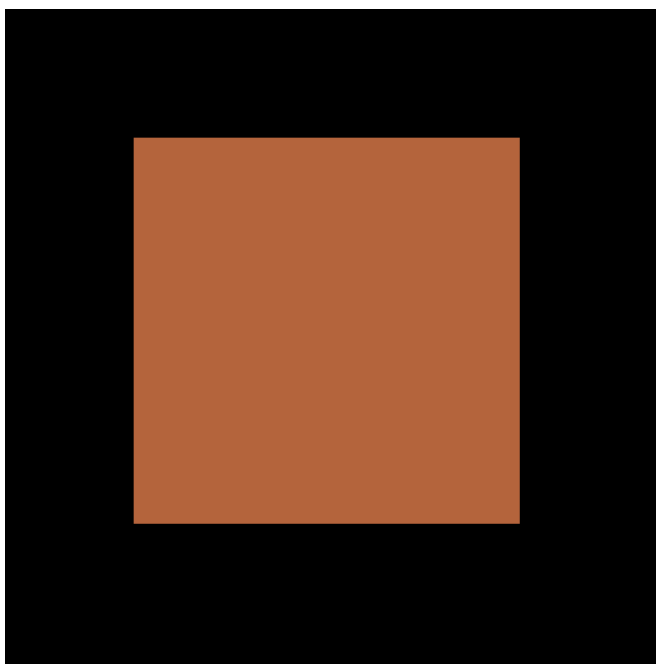
|                    |                                                                          |
|--------------------|--------------------------------------------------------------------------|
| DR Severity Level  | 1 / 4 — Mild DR                                                          |
| Grading Scale      | ETDRS / International Clinical DR Disease Severity Scale                 |
| Model              | EfficientNet-B6 (fine-tuned, 5-class)                                    |
| Recommended Action | Follow-up in 6–12 months. Optimise glycaemic and blood pressure control. |

## ஒளியியல் வட்ட / கோப்பை பிரிவு

வண்ண திறவகை: பச்சை=ஒளியியல் வட்ட | நீலம்=ஒளியியல் கோப்பை | நான்கு பகுதி பகுப்பாய்வு (=மேல், =கீழ், =மலக்கு, =பக்கவாட்டு)



Segmentation (Green=Disc, Blue=Cup)



Overlay with quadrant lines (S/I/N/T)

### Morphometric Measurements

|                              |                                                                                            |
|------------------------------|--------------------------------------------------------------------------------------------|
| Vertical CDR (vCDR)          | 0.450 — CDR within normal limits (< 0.5). Low suspicion for glaucomatous optic neuropathy. |
| Horizontal CDR (hCDR)        | 0.400                                                                                      |
| Area-based CDR               | 0.380                                                                                      |
| Optic Disc — Vertical Ø      | 400 px                                                                                     |
| Optic Disc — Horizontal Ø    | 420 px                                                                                     |
| Optic Cup — Vertical Ø       | 180 px                                                                                     |
| Optic Cup — Horizontal Ø     | 168 px                                                                                     |
| Disc Area (px <sup>2</sup> ) | 125664                                                                                     |
| Cup Area (px <sup>2</sup> )  | 25447                                                                                      |

### Neuroretinal Rim Analysis — ISNT Rule

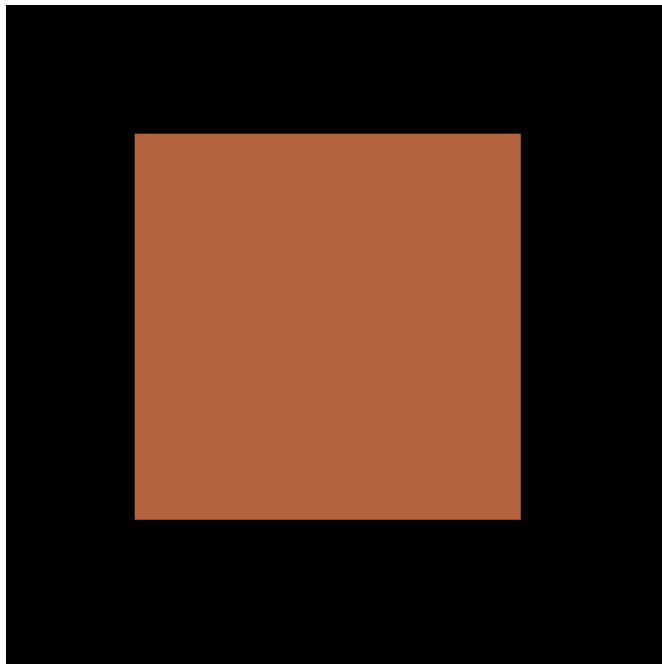
:

|              |                         |
|--------------|-------------------------|
| Superior Rim | 28.2% of total rim area |
| Inferior Rim | 29.1% of total rim area |

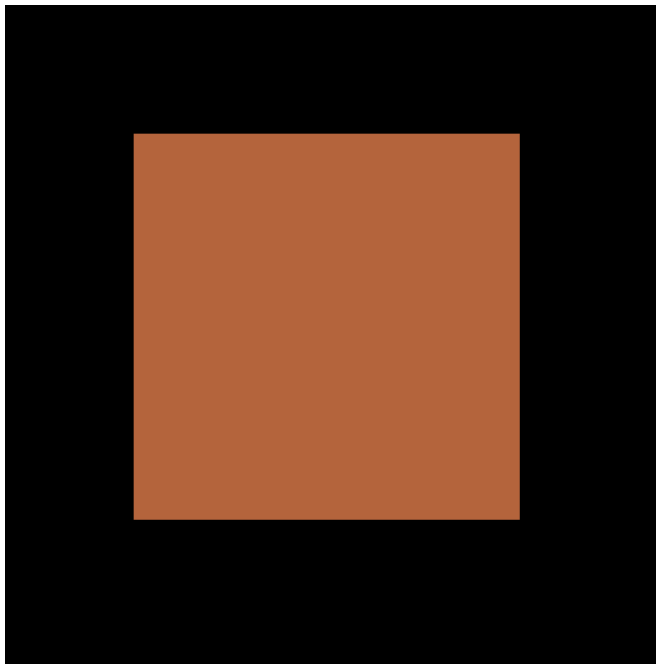
|              |                         |
|--------------|-------------------------|
| Nasal Rim    | 21.4% of total rim area |
| Temporal Rim | 21.3% of total rim area |
| ISNT Rule    | (ISNT )                 |

## புண் கண்டறிதல்

வண்ண திறவகோல்: சிவப்பு=கபின வளையீடுகள் | பச்சை=இரத்தப்போக்கு | நீலம்=நுண்குழிகள் |  
மஞ்சள்=மென்மையான வளையீடுகள்



Lesion segmentation (colour-coded by type)



Overlay on original fundus

### Detected Lesion Areas (pixels<sup>2</sup>)

|                     |                                                           |
|---------------------|-----------------------------------------------------------|
| Hard Exudates (HE)  | 500 px <sup>2</sup> — Lipid deposits from leaking vessels |
| Haemorrhages (HEM)  | 200 px <sup>2</sup> — Intraretinal bleeding               |
| Microaneurysms (MA) | 100 px <sup>2</sup> — Small capillary bulges              |
| Soft Exudates (SE)  | 50 px <sup>2</sup> — Cotton-wool spots                    |
| Model               | UNet + FIAM attention (IDRiD dataset)                     |

## கண் அழுத்த நோய் தரவரிசை

Glaucoma Severity Grade: No Glaucoma

Clinical Interpretation: No evidence of glaucomatous optic neuropathy.

|                     |                                          |
|---------------------|------------------------------------------|
| Model               | SE-ResNet101 (3-class glaucoma grading)  |
| Classification      | No Glaucoma                              |
| Corroborating CDR   | vCDR = 0.450 (suspicion threshold: 0.65) |
| ISNT Rule Violation | No                                       |

## மரத்தாவ சூரக்கம்

|                  |                                             |
|------------------|---------------------------------------------|
| DR Grading       | Mild DR (Level 1/4)                         |
| Glaucoma Grading | No Glaucoma                                 |
| Vertical CDR     | 0.450 — Risk: Low                           |
| ISNT Rule        | Normal                                      |
| Lesion Load      | HE: 500   HEM: 200   SE: 50 px <sup>2</sup> |

## பரிந்துரைகள்

- DR follow-up: Follow-up in 6–12 months. Optimise glycaemic and blood pressure control.

Report ID: A80EB758 | 08 May 2026, 10:12 AM

ஆகி விழித்திரை ஆல் உருவாக்கப்பட்டது

இந்த அறிக்கை மரத்தாவ காற்றிப்பக்காக மட்டுமே ஆகி அமையப்பால் உருவாக்கப்பட்டது. சிகிச்சை மூலம் மூலம் மூலம் அனாதை கண்டிப்புகளும் தகவலாய்ந்த கண் மரத்தாவரால் சரிபார்க்கப்பட வேண்டும்.